Project Seminar 2026





Design of a fuel cell monitoring board in KiCad and development of the operational board software

Lesson 4: Analog circuits and project work

M. Eng. Andreas Hein

E-Mail: andreas.hein@hs-wismar.de

FIW: Fachbereich Elektrotechnik und Informatik

HOCHSCHULE WISMAR
UNIVERSITY OF APPLIED SCIENCES TECHNOLOGY, BUSINESS AND DESIGN

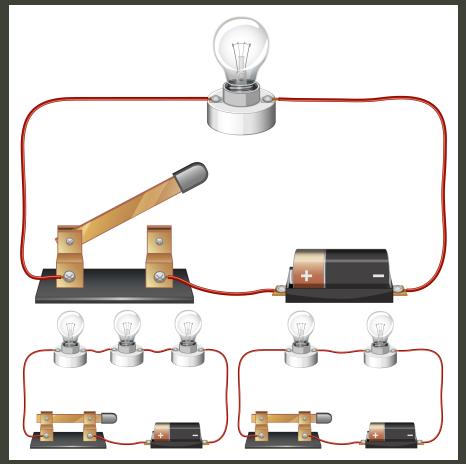
Project Seminar - Project Introduction

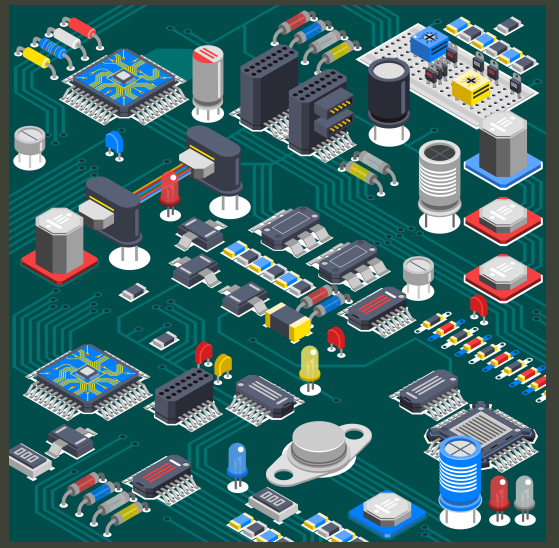






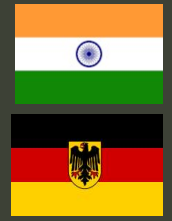
Let's talk about analog circuits ...







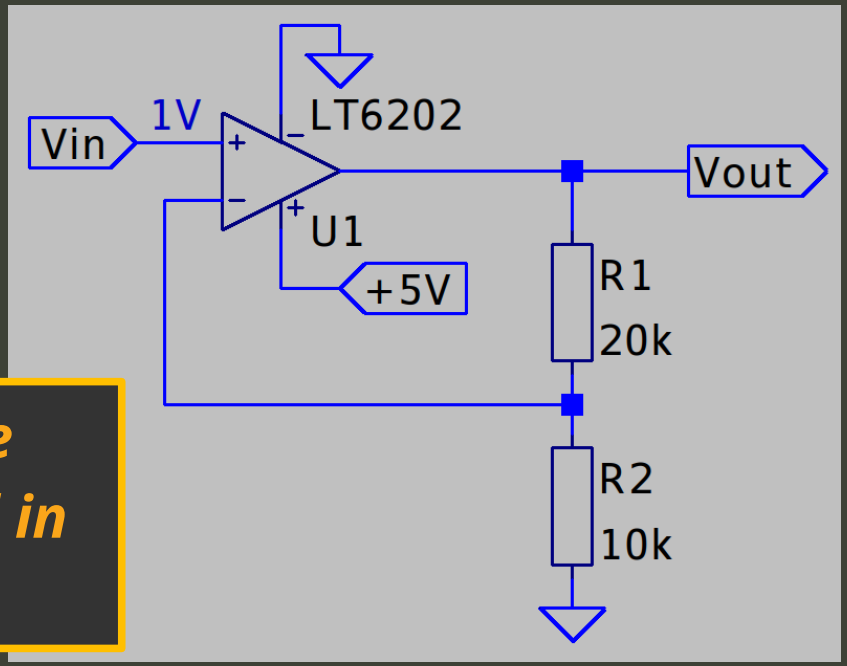
Project Seminar - Project Introduction



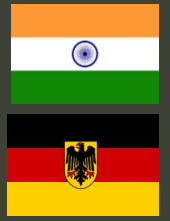
- ***Analog circuits → non-inverting operational amplifier:***
  - LT6202 as non-inverting operational amplifier

→ Characteristics of operational amplifiers are explained in the lecture

→ Calculations to the circuit are presented in the lecture.

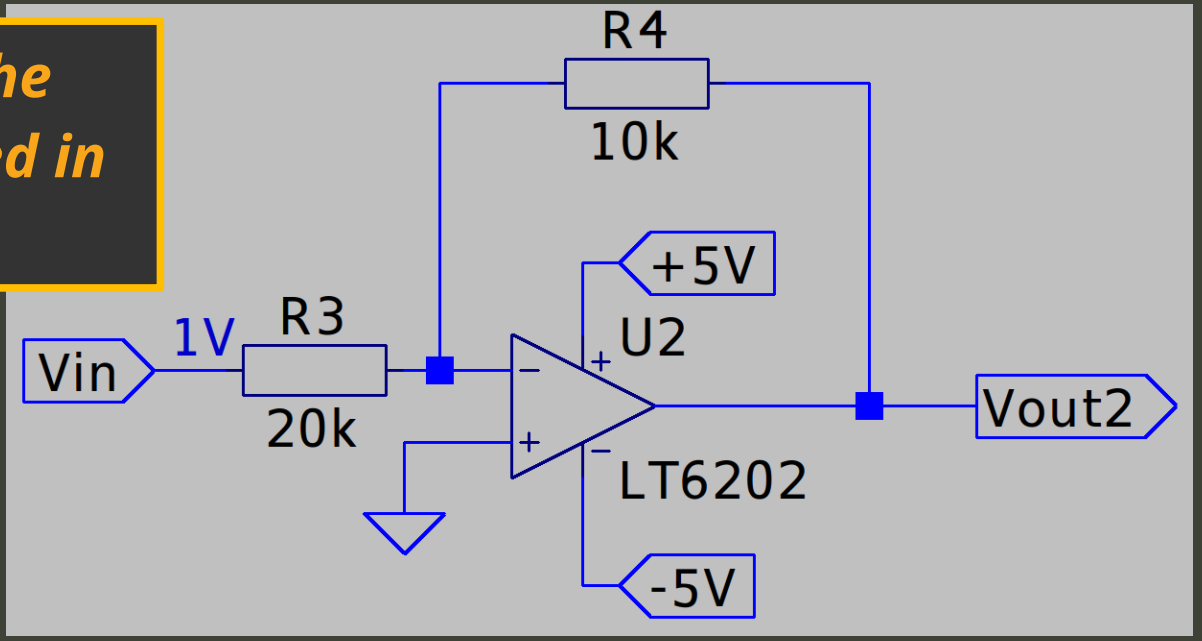


Project Seminar - Project Introduction



- ***Analog circuits → inverting operational amplifier:***
  - LT6202 as inverting operational amplifier

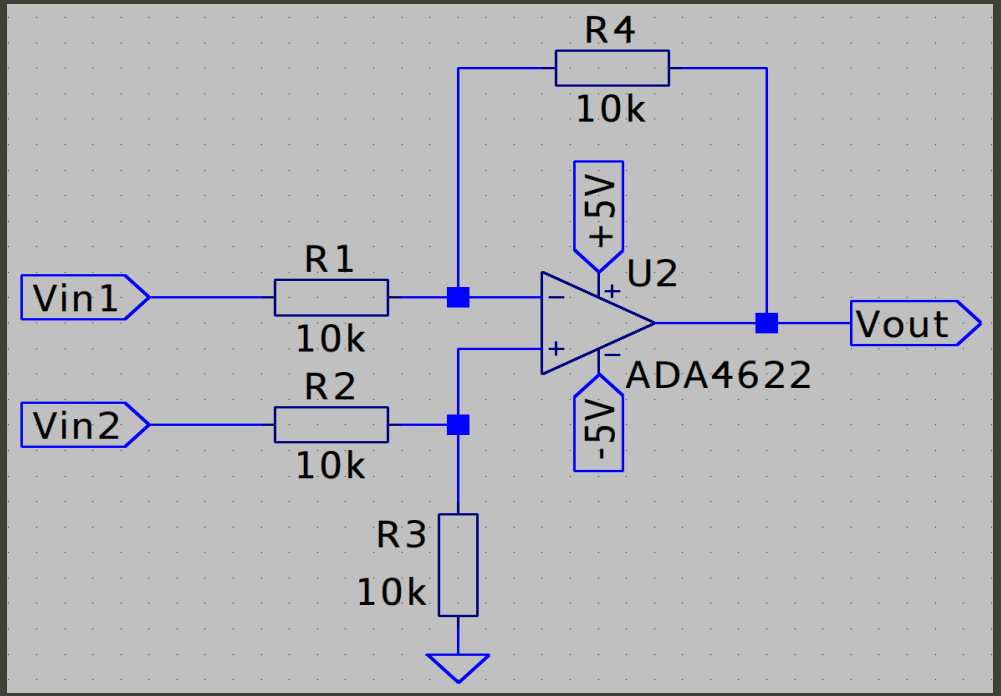
→ Calculations to the circuit are presented in the lecture.



Project Seminar - Project Introduction

- ***Analog circuits → Differential operational amplifier:***
  - ADA4622 as differential operational amplifier

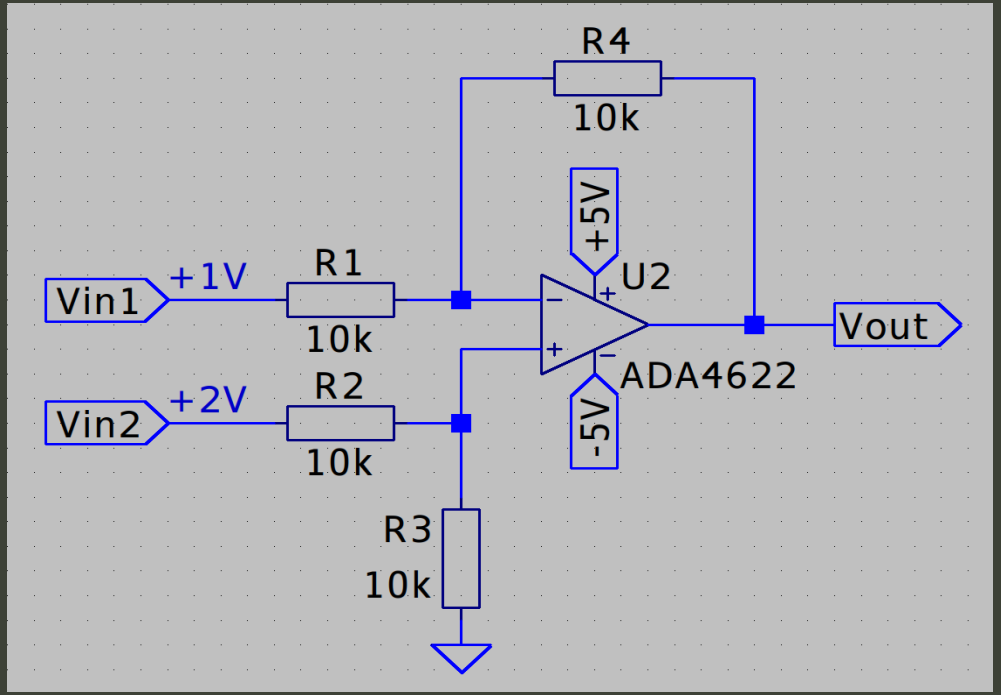
→ Calculations to the circuit are presented in the lecture.



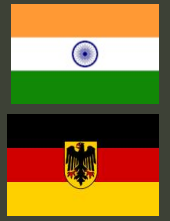
Project Seminar - Project Introduction

- ***Exercise 3: → Differential operational amplifier:***
  - ADA4622 as differential operational amplifier

→ Calculations to the circuit are presented in the lecture.

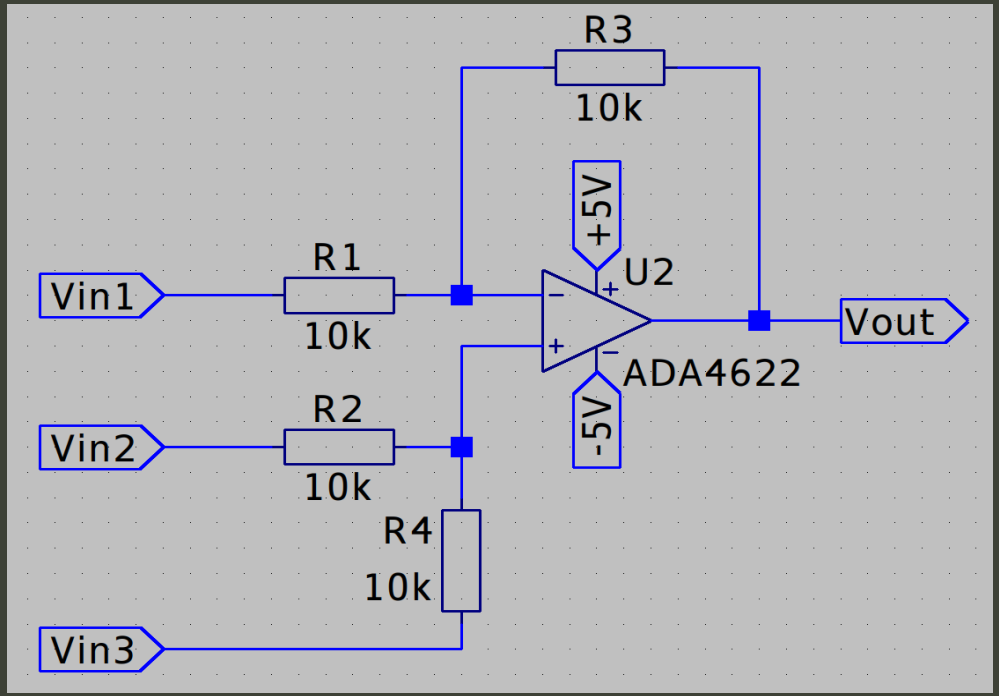


Project Seminar - Project Introduction



- ***Analog circuits → Differential operational amplifier:***
  - ADA4622 as differential operational amplifier with reference voltage

→ Calculations to the circuit are presented in the lecture.



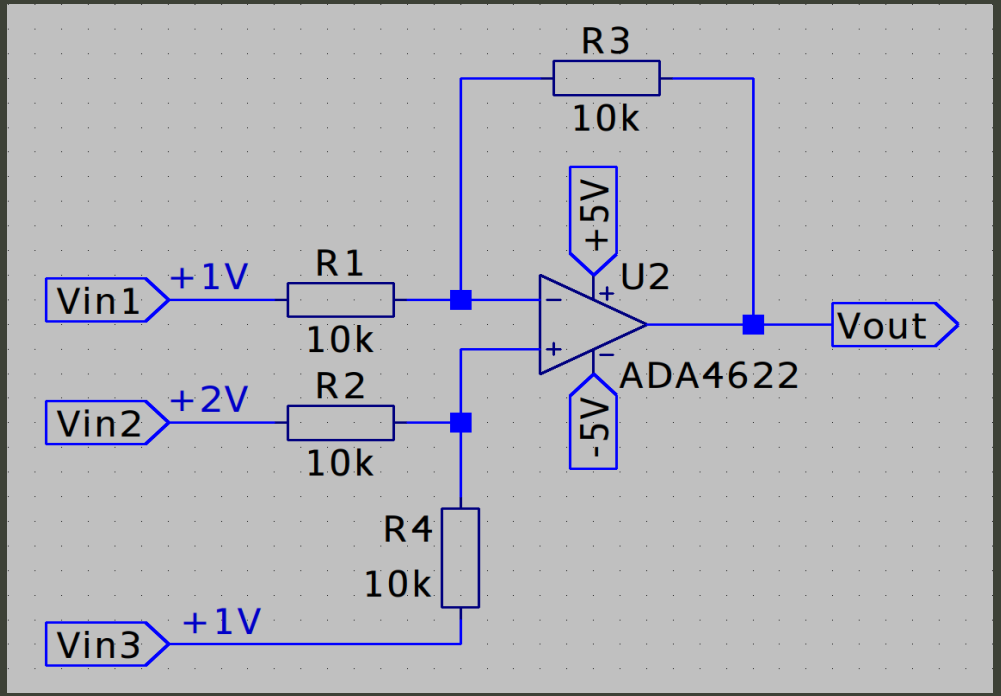


Project Seminar - Project Introduction

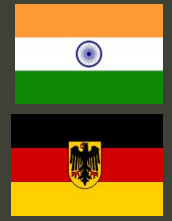


- ***Exercise 4: → Differential operational amplifier:***
  - ADA4622 as differential operational amplifier with reference voltage

→ Calculations to the circuit are presented in the lecture.

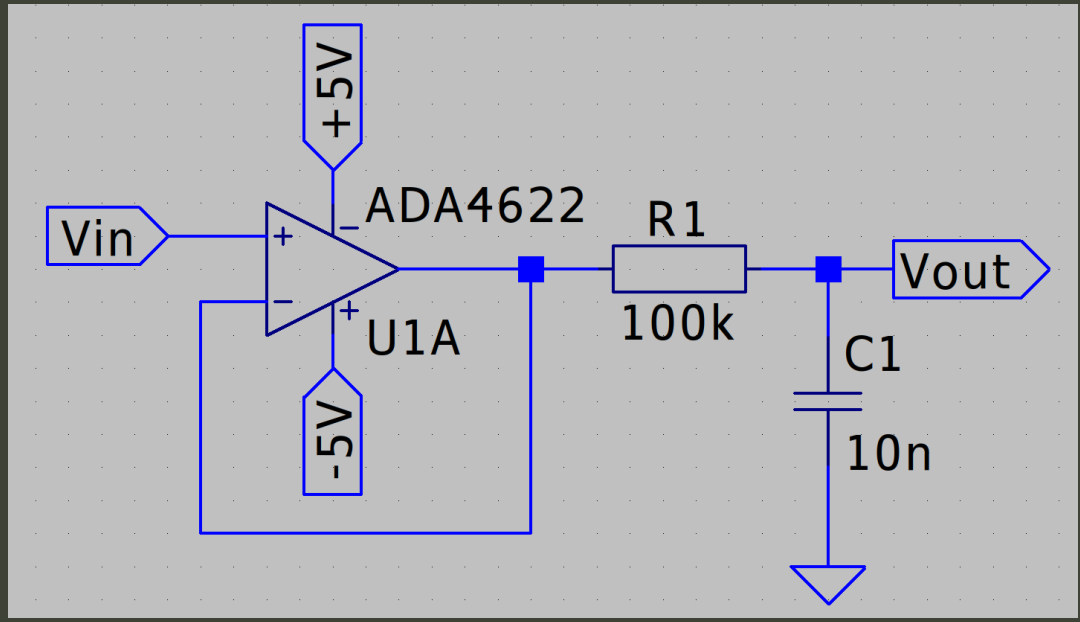


Project Seminar - Project Introduction



- ***Analog circuits → Impedance amplifier → Voltage follower:***
  - ADA4622 as impedance amplifier (voltage follower) with analog low-pass filter

→ Calculations to the circuit are presented in the lecture.



Project Seminar - Project Introduction

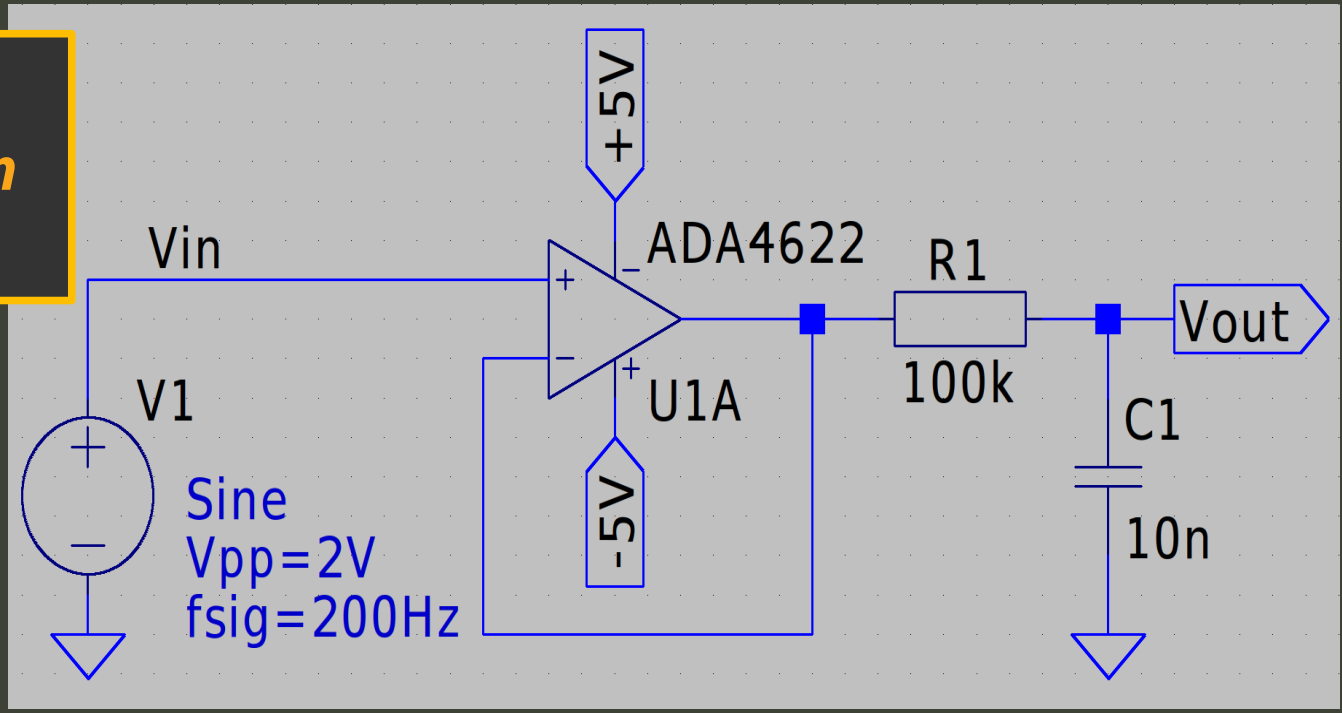
- ***Exercise 5: → Impedance amplifier → Voltage follower:***
  - ADA4622 as impedance amplifier (voltage follower) with analog LPF

→ Calculations to the circuit are presented in the lecture.

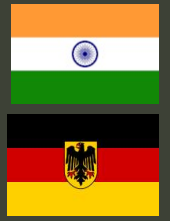
$|\underline{V_u}|(200\,Hz)=?$

a) absolute value ?

b) in dB ?



Project Seminar - Project Introduction

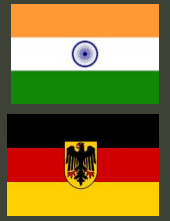


- ***Analog circuits → exercises:***
  - Exercise 6: What is a Hall sensor ? How can a Hall sensor be used to measure the current flowing through a cylindrical copper conductor? Provide a comprehensive written explanation of the measurement principle, including the relevant formulas for calculating the Hall voltage (output voltage of the hall-sensor) !

→ TASK: Answer these questions in your report!



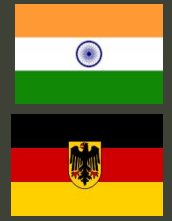
Project Seminar - Project Introduction



- ***Analog circuits & PCB-design → exercises:***
  - Exercise 7: Study the datasheets for the following components and summarize in writing all key information needed to understand how they work, as well as to create the components symbols and the corresponding footprints in KiCad ! :
    → 1. WP-THRBU REDCUBE 74650073
    → 2. RT026xxHBLU Typ 073
    → 3. mpe garry 087 pinhead connector
    → 4. BCS-122-L-S-TE
    → 5. ACS37010LLZATR-030U5
    → 6. MAX49925XATB+T



Project Seminar - Project Introduction



- ***Analog circuits & PCB-design → Exercise 7 continued:***
  - → 7. AD8031ARZ
  - → 8. AD8032ARZ
  - → 9. MAX6070AAUT41+T
  - → 10. MAX6070AAUT21+T
  - → 11. ADS8329IPWR
  - → 12. ADA4099-1BUJZ-RL7
  - → 13. WPME-CDIP 18024115401L
  - → 14. WPME-CDIP 18024215401L
  - → 15. ESP32-S3-DevkitC-1N8R8

→ TASK: Put your solutions in your report!

Project Seminar - SS2026



Project Seminar - Schedule



- ***Lessons and tasks:***
  - 17.04.2026: independent study – repetition & solving exercises
  - until 23.04.2026: Write your own detailed and complete solutions, including step-by-step calculations, for exercises 3 through 7 in your project report!
  - 24.04.2026: Lesson 5 – Working on project → Schematic

… schedule is continously updated!

Project Seminar - SS2026



Thank you !
Have a nice weekend!



